

1. Knowledge type:

① Declarative knowledge: Facts and statements about the world

② Procedural knowledge: Rules, strategies, and sequence of action that can be directly applied to the task

③ Structural knowledge: Relationships and hierarchies between concepts

④ Heuristic knowledge: Rules of thumb or experience based shortcuts.

⑤ Meta-knowledge: know how to use, organize, select or reason with knowledge.



## 2. Knowledge Representation

- ① Logical KR: clear rules or statements that are either true or false  
Technique: Propositional logic, First-order logic
- ② Structured KR: Organized knowledge into structured formats like hierarchies, graphs or frames
- ③ Probabilistic KR: Represent knowledge where uncertainty exists by assigning probabilities to outcomes.
- ④ Represent knowledge as vectors in a high dimensional space.



### 3. Reasoning

- ① Deductive: General rule  $\rightarrow$  Specific conclusion
  - ② Inductive: Specific observation  $\rightarrow$  General Conclusion
  - ③ Abductive: Incomplete observation  $\rightarrow$  Best Explanation
- 

1. A single propositional symbol can only represent an atomic proposition.

2. Syntax: ①  $\neg$  (not): negation    ②  $\wedge$  (and): conjunction  
③  $\vee$  (or): disjunction    ④  $\Rightarrow$  (implies)    ⑤  $\Leftrightarrow$  (if and only if): biconditional

3. Precedence:  $\underbrace{() \neg \wedge \vee \Rightarrow \Leftrightarrow}_{\text{Highest} \quad \quad \quad \text{Lowest}} \rightarrow$



#### 4. Well-formed formula (WFF)

- ① Any proposition symbol is a WFF
- ② If  $\alpha$  is a WFF,  $\neg \alpha$  is a WFF
- ③ If  $\alpha$  and  $\beta$  are WFFs, the followings are also

WFFs:

$(\alpha \wedge \beta)$ ,  $(\alpha \vee \beta)$ ,  $(\alpha \Rightarrow \beta)$ ,  $(\alpha \Leftrightarrow \beta)$

5. Model: An assignment of truth value to every proposition symbol

6. Entailment:  $\beta \models \alpha$  if and only if  $M(\beta) \subseteq M(\alpha)$   
(i.e. If  $\beta$  is true, then  $\alpha$  must be true)



7. Tautology: WFF that is always true

8. Contradiction: A WFF that is always false

9. Satisfiability: A WFF that is true / satisfied by at least 1 model

10. Knowledge base (KB): A set of WFFs that represent known facts or rules about the world.

11. Model-checking algorithm

① Enumerate all the possible models

② Check if  $\phi$  is true in every model in which KB is



true

$\frac{\alpha_1, \alpha_2, \dots, \alpha_k}{\beta} \longrightarrow \text{Premise}$   
 $\beta \longrightarrow \text{Conclusion}$

12. Inference Rule:

① Modus Ponens

$$\frac{\alpha \Rightarrow \beta, \alpha}{\beta}$$

② And Elimination

$$\frac{\alpha \wedge \beta}{\alpha}$$

③ Logical Equivalence:  $\alpha \equiv \beta \Rightarrow$

④ Resolution Rule:  $\frac{\alpha \vee \beta, \neg \alpha}{\beta}$

$$\frac{\frac{\alpha}{\beta} \quad \frac{\beta}{\alpha} \quad \alpha_1 \vee \dots \vee \alpha_n, \gamma, \vee \dots \vee \gamma_m}{\alpha_1 \vee \dots \vee \alpha_n \vee \gamma, \vee \dots \vee \gamma_m}$$

13. Resolution Algorithm: check if  $K\beta \models \alpha \Rightarrow$   
check if  $K\beta \wedge \neg \alpha$  is a contradiction



# Propositional Logic : Inference

## ➤ Forward Chaining Algorithm

Step 1: Start with atomic propositions in the KB

Step 2: Apply inference rules to infer new facts and add them to the KB

Step 3: Continue until: -The goal (query) is derived, or  
-No more new facts can be inferred

e.g.,  $KB = \{A, B, A \wedge B \Rightarrow C, A \wedge P \Rightarrow L, B \wedge C \Rightarrow P\}$ , Query: L

Step 1: A is true. B is true

Step 2: From  $A \wedge B \Rightarrow C$ , infer C is true.

Update  $KB = \{A, B, A \wedge B \Rightarrow C, A \wedge P \Rightarrow L, B \wedge C \Rightarrow P, C\}$

From  $B \wedge C \Rightarrow P$ , infer P is true.

Update  $KB = \{A, B, A \wedge B \Rightarrow C, A \wedge P \Rightarrow L, B \wedge C \Rightarrow P, C, P\}$

From  $A \wedge P \Rightarrow L$ , infer L is true. ➡



# Propositional Logic : Inference

## ➤ Backward Chaining Algorithm

Step 1: Find implications in the KB whose conclusions are query.

Step 2: Try to prove the premises of those implications.

Step 3: Recursively prove subgoals until reaching the atomic propositions in the KB

e.g.,  $KB = \{A, B, A \wedge B \Rightarrow C, A \wedge P \Rightarrow L, B \wedge C \Rightarrow P\}$ , Query: L

Step 1: Find  $A \wedge P \Rightarrow L$  whose conclusion is L, check if  $A \wedge P$  is true.

Step 2: A is true; Find  $B \wedge C \Rightarrow P$  whose conclusion is P, check if  $B \wedge C$  is true.

B is true; Find  $A \wedge B \Rightarrow C$  whose conclusion is C

$A \wedge B$  is true ➡



# Propositional Logic : Inference

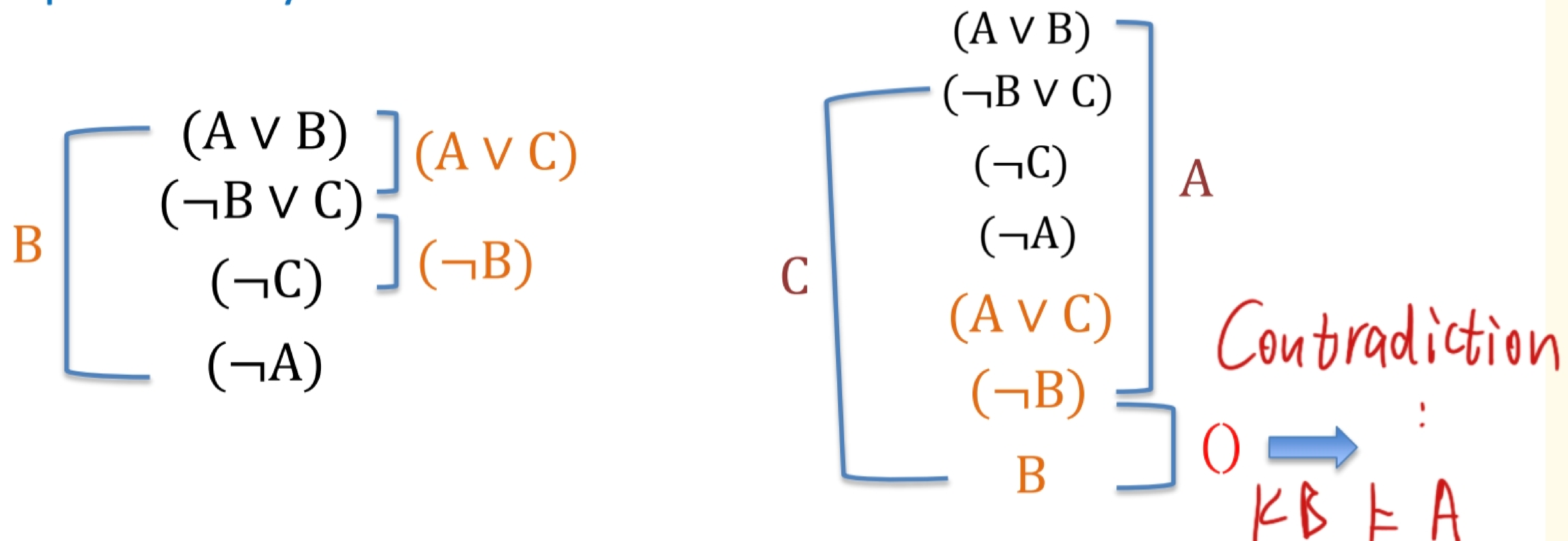
## ➤ Resolution Algorithm

e.g., does  $KB = \{A \vee B, \neg B \vee C, \neg C\}$  entail  $A$ ?

Step 1. Convert  $KB \wedge \neg A$  into CNF

$$KB \wedge \neg A = (A \vee B) \wedge (\neg B \vee C) \wedge (\neg C) \wedge (\neg A)$$

Step 2. Apply resolution rule to each pair of clauses that contains complementary literals





## ➤ Logical Equivalence

Two WWFs  $\alpha$  and  $\beta$  are logically equivalent if they are true in the same set of models.

$\alpha \equiv \beta$  if and only if  $\alpha \models \beta$  and  $\beta \models \alpha$

$$(\alpha \wedge \beta) \equiv (\beta \wedge \alpha) \quad \text{commutativity of } \wedge$$

$$(\alpha \vee \beta) \equiv (\beta \vee \alpha) \quad \text{commutativity of } \vee$$

$$((\alpha \wedge \beta) \wedge \gamma) \equiv (\alpha \wedge (\beta \wedge \gamma)) \quad \text{associativity of } \wedge$$

$$((\alpha \vee \beta) \vee \gamma) \equiv (\alpha \vee (\beta \vee \gamma)) \quad \text{associativity of } \vee$$

$$\neg(\neg\alpha) \equiv \alpha \quad \text{double-negation elimination}$$

$$(\alpha \Rightarrow \beta) \equiv (\neg\beta \Rightarrow \neg\alpha) \quad \text{contraposition}$$

$$(\alpha \Rightarrow \beta) \equiv (\neg\alpha \vee \beta) \quad \text{implication elimination}$$

$$(\alpha \Leftrightarrow \beta) \equiv ((\alpha \Rightarrow \beta) \wedge (\beta \Rightarrow \alpha)) \quad \text{biconditional elimination}$$

$$\neg(\alpha \wedge \beta) \equiv (\neg\alpha \vee \neg\beta) \quad \text{De Morgan}$$

$$\neg(\alpha \vee \beta) \equiv (\neg\alpha \wedge \neg\beta) \quad \text{De Morgan}$$

$$(\alpha \wedge (\beta \vee \gamma)) \equiv ((\alpha \wedge \beta) \vee (\alpha \wedge \gamma)) \quad \text{distributivity of } \wedge \text{ over } \vee$$

$$(\alpha \vee (\beta \wedge \gamma)) \equiv ((\alpha \vee \beta) \wedge (\alpha \vee \gamma)) \quad \text{distributivity of } \vee \text{ over } \wedge$$



# Propositional Logic: Semantics

## ➤ Truth Table

A table that shows the **truth value** of a WWF for **every possible combination** of truth values of its atomic propositions

| P     | Q     | $\neg P$ | $P \wedge Q$ | $P \vee Q$ | $P \Rightarrow Q$ | $P \Leftrightarrow Q$ |
|-------|-------|----------|--------------|------------|-------------------|-----------------------|
| true  | true  | false    | true         | true       | true              | true                  |
| true  | false | false    | false        | true       | false             | false                 |
| false | true  | true     | false        | true       | true              | false                 |
| false | false | true     | false        | false      | true              | true                  |